Statistics Report: Heart Disease Classification

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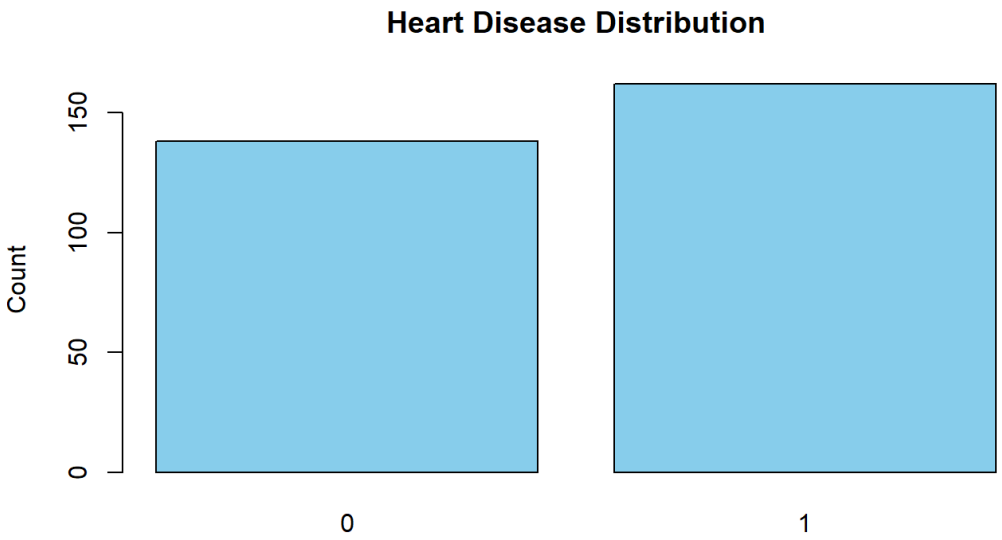
1. Introduction

The objective of this analysis is to investigate factors associated with heart disease and to build classification models to predict disease status. The dataset contains demographic, clinical, and diagnostic variables, with a binary response variable indicating the presence (1) or absence (0) of heart disease.

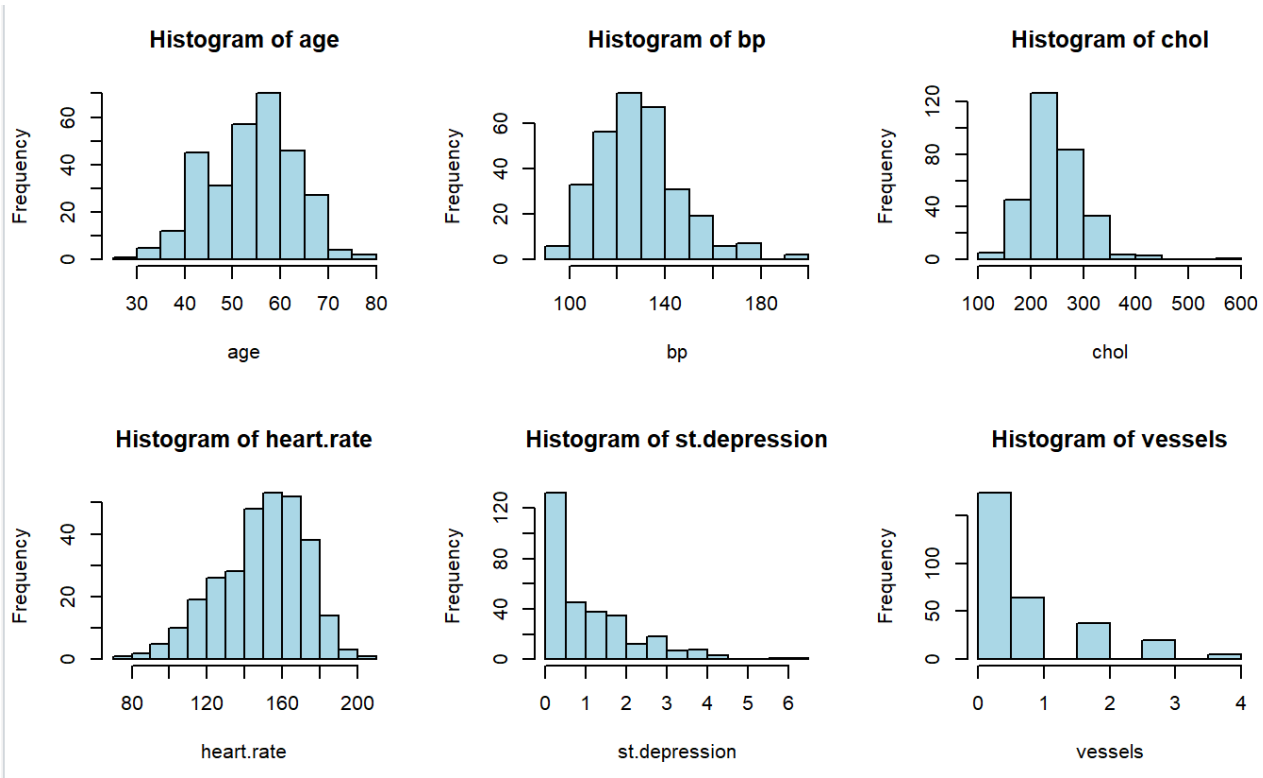
2. Variable Analysis

2.1 Response Variable



The bar plot shows that the dataset is relatively balanced, with slightly more observations in the diseased group (0) than the non-diseased group (1).

2.2 Distribution of Numerical Variables

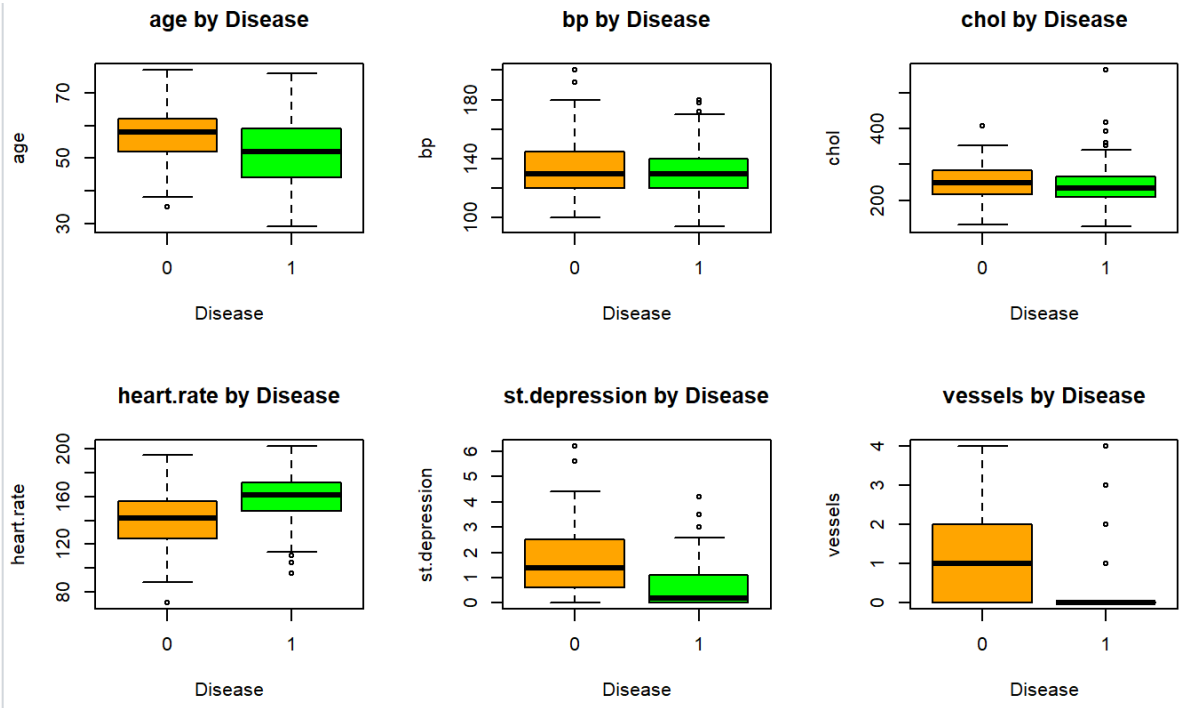


From the histograms:

- **Age**: Approximately bell-shaped, centred around 50–60 years.
- **Blood Pressure (bp)**: Slight right skew, with most values between 110–150.
- **Cholesterol (chol)**: Right-skewed with some extreme high values (outliers).
- **Heart Rate**: Roughly symmetric, centred around 140–160.
- **ST Depression**: Strong right skew, with many values close to 0.
- **Vessels**: Highly discrete and skewed (mostly 0 or 1).

Interpretation:
Several variables (chol, st.depression, vessels) exhibit skewness and outliers, which may influence model performance.

2.3 Boxplots by Disease Status



The boxplots reveal key differences:

- **Heart Rate**: Higher median for diseased patients.
- **ST Depression**: Lower median for diseased patients (interesting inverse pattern).
- **Vessels**: Strong separation. Diseased individuals tend to have more vessels involved.
- **Age, BP, Chol**: Differences exist but are less pronounced.

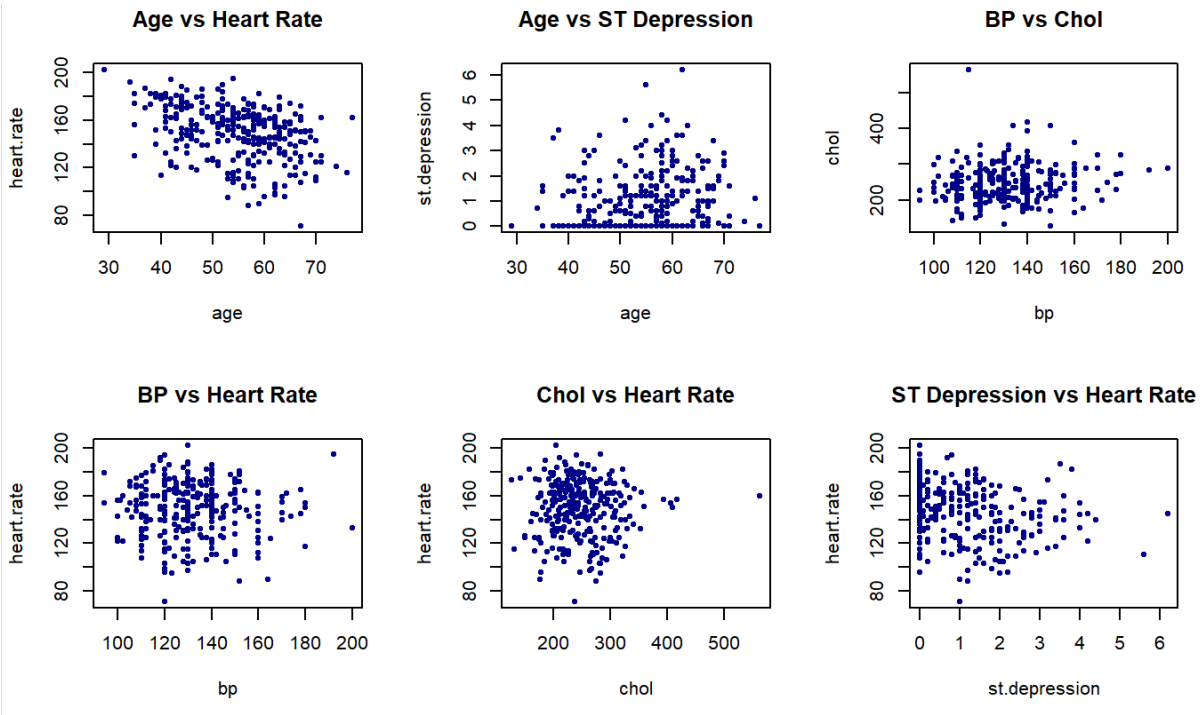
Interpretation:

Vessels, **heart.rate** and **st.depression** appear to be strong predictors.

The boxplots indicate that **vessels** is a strongest predictor, showing clear separation between disease groups. **Heart rate** also demonstrates a noticeable upward shift in the diseased group, suggesting a positive association with heart disease. In contrast, variables such as age, blood pressure, and cholesterol exhibit substantial overlap between groups, indicating weaker individual predictive power.

ST depression displays an inverse pattern, suggesting a more complex relationship that may be better captured by non-linear models.

2.4 Scatterplot Relationships

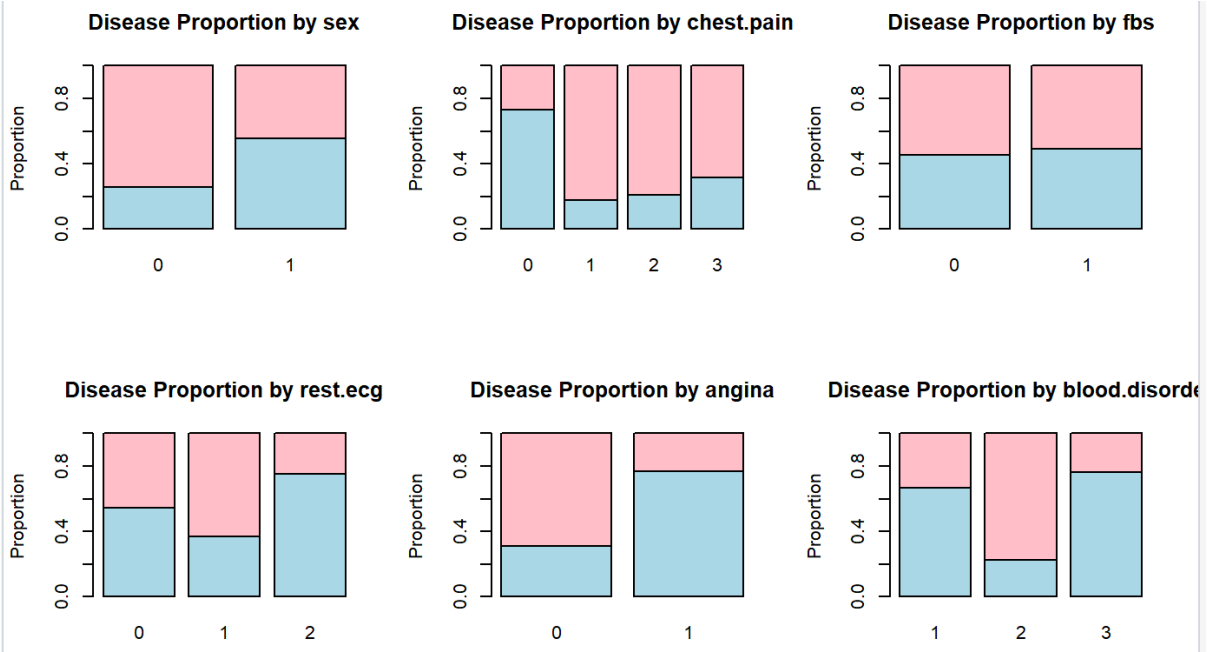


From the scatterplots:

- **Age vs Heart Rate**: Slight negative relationship.
- **ST Depression vs Heart Rate**: Weak negative trend.
- **BP vs Chol / Heart Rate**: No strong linear relationship.

Interpretation:
The scatterplots indicate weak and non-linear relationships between predictors, with no clear linear trends observed. This suggests that linear models alone may not fully capture the structure of the data, and supports the use of non-linear methods such as KNN and decision trees, which can model complex and threshold-based relationships.

2.5 Categorical Variables vs Disease



From the proportion bar plots:

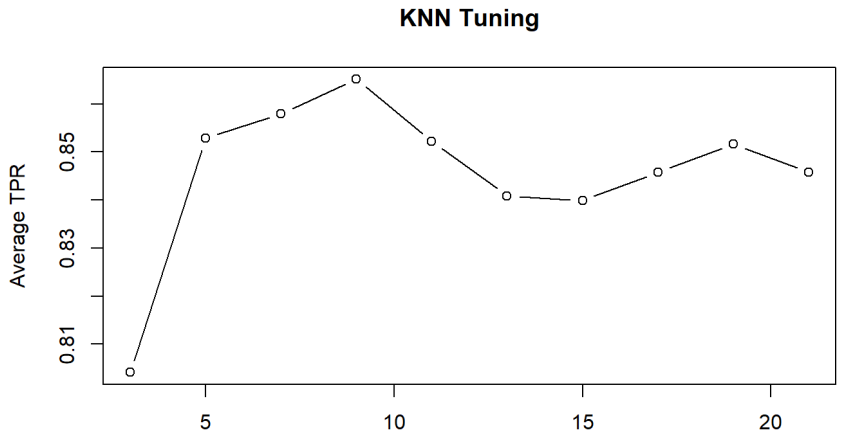
- **Chest Pain**: Strong variation across categories. An important predictor.
- **Angina**: Clear separation between disease groups.
- **Blood Disorder**: Highly informative.
- **Sex, FBS, ECG**: Some differences but weaker.

Interpretation:

Variables such as **chest pain**, **angina**, and **blood disorder** exhibit substantial differences in disease proportions across categories, indicating strong associations with the response variable. In contrast, variables such as **sex, fasting blood sugar, and ECG** show more similar distributions, suggesting weaker predictive power.

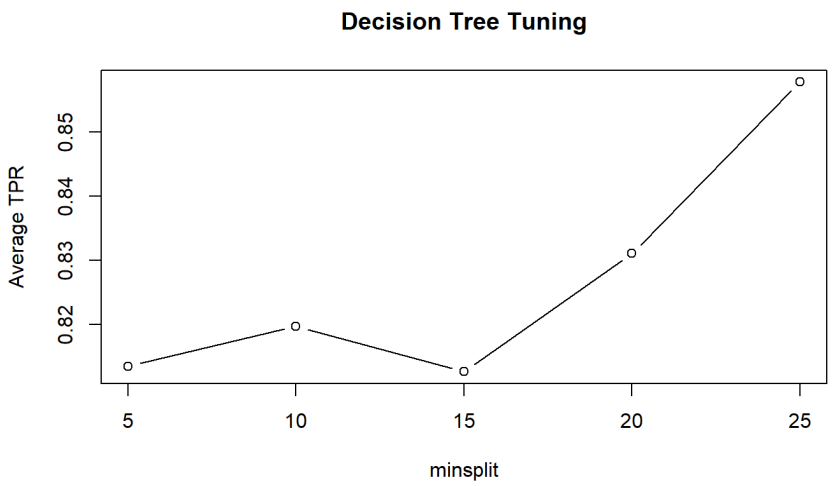
3. Model Development

3.1 KNN Tuning

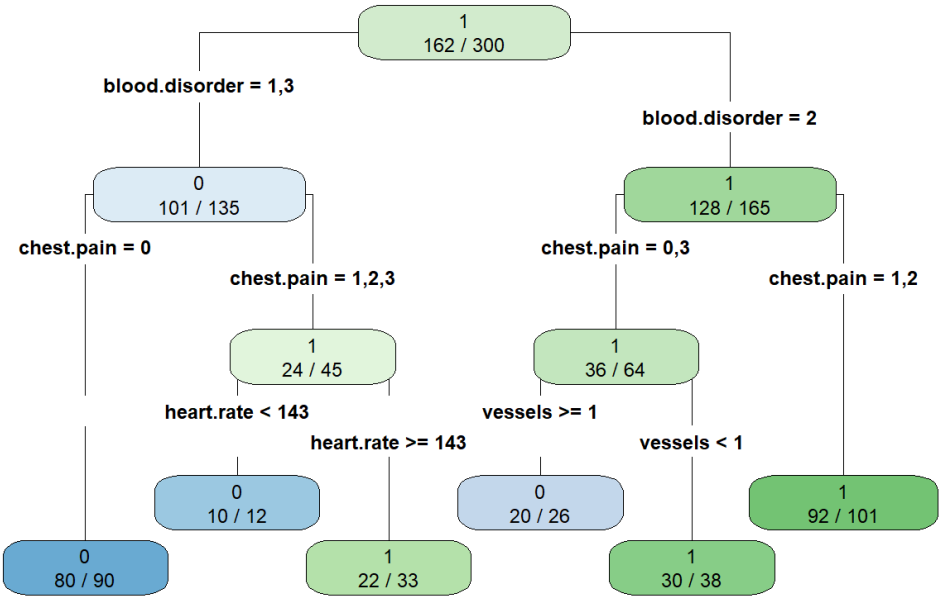


k = 9 gives the highest True Positive Rate (TPR)

3.2 Decision Tree



Decision tree tuning was performed by varying the minsplit parameter and evaluating performance using 5-fold cross-validation. The true positive rate (TPR) increased as minsplit increased from 5 to approximately 15, indicating reduced overfitting and improved generalisation. Beyond this point, TPR declined slightly, suggesting that the model became too simple and began to underfit. Therefore, minsplit = 15 was selected as the optimal value.



The decision tree identifies blood disorder as the root node, indicating it is the most important predictor of heart disease. Subsequent splits on chest pain further refine classification, followed by vessels and heart rate, which provide additional discrimination within subgroups. This hierarchical structure suggests that heart disease risk is driven by a combination of categorical and threshold-based relationships, supporting the presence of non-linear interactions among predictors.

4. Model Development

4.1 Confusion Matrices

Model	True Negative	False Positive	False Negative	True Positive
KNN	109	29	20	142
Decision Tree	110	28	18	144
Logistic Regression	109	29	14	148
Naive Bayes	112	26	20	142

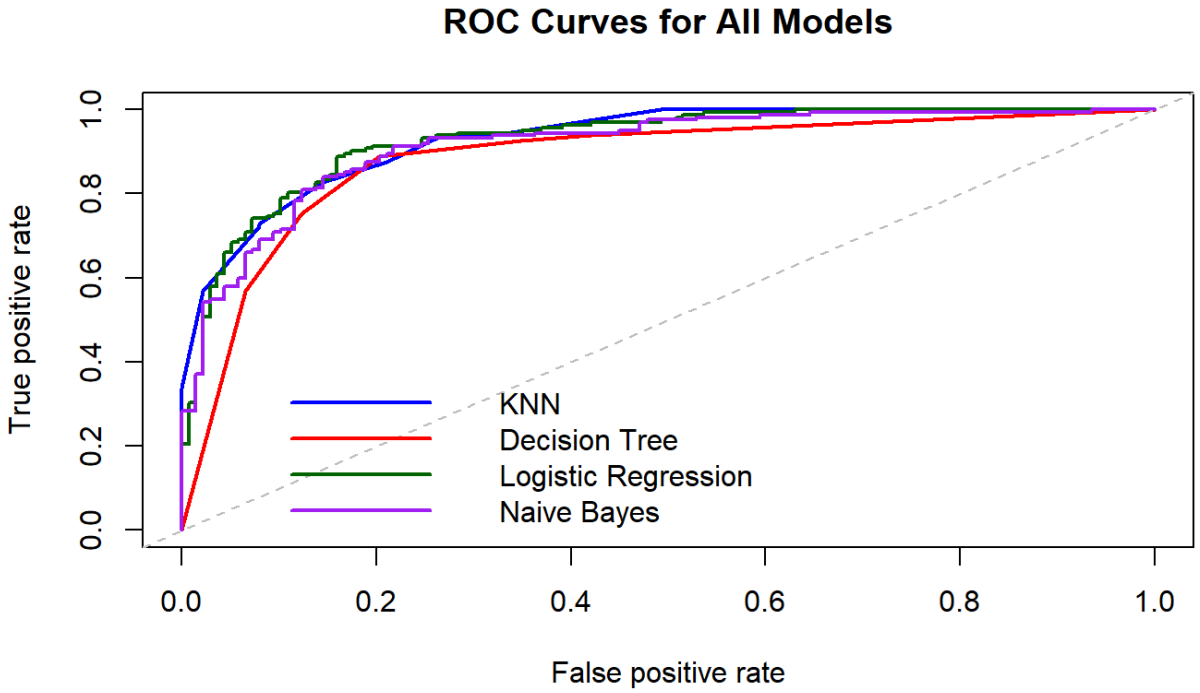
Finding the balance:

- Logistic Regression:
  - Minimises False Negatives (sensitivity prioritised)
- Naive Bayes:
  - Minimises False Positives (precision prioritised)

4.2 Performance of Models

Model	True Positive Rate (TPR)	Precision	AUC
KNN	0.8765	0.8304	0.9298
Decision Tree	0.8889	0.8372	0.8831
Logistic Regression	0.9136	0.8362	0.9273
Naive Bayes	0.8765	0.8452	0.9138

4.3 ROC Curve



5. Insights

The results reveal several important insights into both the data structure and model behaviour. Despite the exploratory analysis indicating weak and non-linear relationships between individual predictors and the response variable, logistic regression performs the best overall. This can be explained by its ability to combine multiple predictors additively, allowing even

weak individual signals to contribute meaningfully when considered together. In contrast, decision trees rely on a limited number of hierarchical splits, which, while interpretable, may overlook subtle interactions and cumulative effects across variables. This explains the slightly lower AUC observed for the decision tree model.

KNN achieves the highest AUC, indicating strong overall discrimination between classes. However, its slightly lower TPR compared to logistic regression suggests that while it is effective at ranking observations by risk, it is less optimal when using a fixed classification threshold. This highlights an important distinction between ranking performance and classification performance.

Naive Bayes also performs competitively despite its strong assumption of feature independence. This is likely because the dataset contains several highly informative predictors, such as vessels, chest pain, and blood disorder, which dominate the classification process. As a result, the violation of the independence assumption does not significantly degrade performance.

Overall, the dataset appears to contain a mix of strong threshold-based predictors and weaker continuous variables. This combination allows both linear and non-linear models to perform well, but favours models that can effectively aggregate information across multiple features.

6. Conclusion

This analysis identifies key predictors of heart disease including:

- Blood disorder
- Chest pain
- Vessels
- Heart rate

which demonstrate strong associations with the response variable. Variables such as age, blood pressure, and cholesterol exhibit weaker individual effects but still contribute when combined with other predictors.

Among the models evaluated, logistic regression emerges as the most suitable model for this task. It achieves the highest true positive rate, indicating superior ability to correctly identify patients with heart disease, while also maintaining strong overall discriminatory performance as reflected by its high AUC. Although KNN demonstrates slightly higher AUC and Naive Bayes achieves the highest precision, these advantages are secondary in the context of healthcare decision-making.

In medical applications, the cost of misclassification is asymmetric. A false negative can lead to missed diagnosis, delayed treatment, and potentially severe health consequences. In contrast, a false positive typically results in additional testing or precautionary treatment, which, while undesirable, is significantly less harmful. Therefore, prioritising models with low false negative rates and high true positive rates (sensitivity) is critical.

Logistic regression satisfies this requirement by producing the lowest number of false negatives and the highest TPR among all models. This makes it the most appropriate choice for deployment in a clinical screening context, where the primary objective is to ensure that as many true cases as possible are detected.

In conclusion, while multiple models demonstrate strong performance, the healthcare context necessitates prioritising sensitivity over other metrics. Logistic regression is therefore recommended as the final model, as it provides the best balance between predictive performance and clinical risk considerations.